

Visitor Management Backend - Technical Documentation

Spring Boot 4.1.0 · Java 21 · PostgreSQL 17 · Hibernate 7.4

1. Architecture Overview

+-----+ REST Controllers +-----+	
TimeSlotController · VisitorController	
RegistrationController · BriefingController	
EventController · FeedbackController	
EducationalProgramController · BookingController	
+-----+ Services +-----+	
SchedulingServiceImpl · RegistrationServiceImpl	
EventServiceImpl · FeedbackServiceImpl	
EducationalProgramServiceImpl · BookingServiceImpl	
BookingReminderJob (rappel 24h)	
SchedulingRules (constants)	
+-----+ Repositories (JPA) +-----+	
TimeSlotRepository · VisitorRepository	
RegistrationRepository · BriefingRepository	
EventRepository · FeedbackRepository	
SurveySendRepository · TourStopRepository	
WorkshopRepository · AgriActivityRepository	
BookingRepository	
+-----+ JPA Entities +-----+	
TimeSlot · Visitor · Registration · Briefing · Event	
Feedback · SurveySend · TourStop · Workshop	
AgriActivity · Booking	
(extend BaseEntity: id + createdAt + updatedAt)	
+-----+ core/ - shared infrastructure +-----+	
BaseEntity · GlobalExceptionHandler · ApiError	
ResourceNotFoundException · ConflictException	
BusinessException · NotificationService (console/SMTP)	
WebConfig (CORS) · DevDataSeeder (dev profile only)	

Packages:

- core/ - shared infra only (BaseEntity, exceptions, handler, notification, CORS config, dev seeder)
- modules/visitormanagement/ - Scheduling + Registration sub-modules
- Enums live in the module package (not core/)

2. Database Schema

Entity-Relationship Diagram

+-----+ time_slot +-----+		+-----+ visitor +-----+	
id BIGINT PK		id BIGINT PK	
slot_date DATE NOT NULL		full_name VARCHAR(150)	NOT NULL
start_time TIME NOT NULL		group_size INT	NOT NULL, DEFAULT 1
end_time TIME NOT NULL		email VARCHAR(200)	UNIQUE
max_capacity INT NOT NULL		phone VARCHAR(30)	
status VARCHAR(20) NOT NULL		language VARCHAR(40)	
guide_id BIGINT		visitor_type VARCHAR(20)	NOT NULL

```

| created_at INSTANT | NOT NULL | special_needs VARCHAR(500) |
| updated_at INSTANT | NOT NULL | created_at INSTANT | NOT NULL
+-----+-----+
| 1 | 1 | |
| 1 | 1 |
| * | 1 | * |
+-----+-----+
| registration |
+-----+-----+
| id BIGINT | PK |
| visitor_id BIGINT | FK -> visitor (NOT NULL) |
| time_slot_id BIGINT | FK -> time_slot (nullable, NULL for |
| | event-only registrations) |
| event_id BIGINT |
| visit_purpose VARCHAR(20) | NOT NULL |
| is_prospect BOOLEAN | NOT NULL, DEFAULT false |
| status VARCHAR(20) | NOT NULL, DEFAULT 'PENDING' |
| created_at INSTANT | NOT NULL |
| updated_at INSTANT | NOT NULL |
| UNIQUE (visitor_id, time_slot_id) |
+-----+-----+
| 1 |
| 1 |
+-----+-----+
| briefing |
+-----+-----+
| id BIGINT | PK |
| registration_id | FK -> registration |
| | (NOT NULL, UNIQUE) |
| delivered_at INSTANT |
| staff_member VARCHAR(150) |
| signature VARCHAR(500) |
| status VARCHAR(20) | NOT NULL, 'PENDING' |
| created_at INSTANT | NOT NULL |
| updated_at INSTANT | NOT NULL |
+-----+-----+

```

registration.event_id -> event.id (no FK constraint, soft link)

```

+-----+-----+
| event |
+-----+-----+
| id BIGINT | PK |
| title VARCHAR(150) | NOT NULL |
| event_type VARCHAR(30) | NOT NULL |
| start_date_time TIMESTAMP | NOT NULL |
| end_date_time TIMESTAMP | NOT NULL |
| max_capacity INT | NOT NULL, DEFAULT 10 |
| status VARCHAR(20) | NOT NULL, 'DRAFT' |
| description VARCHAR(1000) |
| location VARCHAR(150) |
| created_at INSTANT | NOT NULL |
| updated_at INSTANT | NOT NULL |
+-----+-----+

+-----+-----+
| feedback |
+-----+-----+
| id BIGINT | PK |
| visitor_id BIGINT | FK -> visitor (NOT NULL) |
| survey_send_id BIGINT | FK -> survey_send (nullable) |

```

```

| origin    VARCHAR(20)| NOT NULL    |
| rating     INT| NOT NULL (1..5)    |
| briefing_clear VARCHAR(1000)|
| educational_value VARCHAR(1000)|
| recommend   VARCHAR(1000)|
| comment     VARCHAR(1000)|
| routed_to   VARCHAR(60)|
| submitted_at INSTANT| NOT NULL    |
| created_at   INSTANT| NOT NULL    |
| updated_at   INSTANT| NOT NULL    |
+-----+

+-----+
|          survey_send          |
+-----+
| id            BIGINT| PK          |
| visitor_id     BIGINT| FK -> visitor (NOT NULL)|
| channel        VARCHAR(20)| NOT NULL    |
| message_template VARCHAR(500)|
| sent_at        INSTANT| NOT NULL    |
| status         VARCHAR(20)| NOT NULL, 'SENT'|
| created_at     INSTANT| NOT NULL    |
| updated_at     INSTANT| NOT NULL    |
+-----+

+-----+
|          tour_stop          |
+-----+
| id            BIGINT| PK          |
| name          VARCHAR(150)| NOT NULL    |
| position       INT| NOT NULL    |
| description    VARCHAR(2000)|
| duration_minutes INT| NOT NULL    |
| max_capacity   INT|
| location      VARCHAR(150)|
| demo          VARCHAR(1000)|
| safety_notes  VARCHAR(1000)|
| active        BOOLEAN| NOT NULL, true |
| created_at     INSTANT| NOT NULL    |
| updated_at     INSTANT| NOT NULL    |
+-----+

+-----+
|          workshop          |
+-----+
| id            BIGINT| PK          |
| name          VARCHAR(150)| NOT NULL    |
| duration_minutes INT| NOT NULL    |
| target_group  VARCHAR(60)| NOT NULL    |
| facilitator   VARCHAR(100)|
| description   VARCHAR(1000)|
| status        VARCHAR(20)| NOT NULL, 'DRAFT'|
| created_at     INSTANT| NOT NULL    |
| updated_at     INSTANT| NOT NULL    |
+-----+

+-----+
|          agri_activity          |
+-----+
| id            BIGINT| PK          |
| name          VARCHAR(150)| NOT NULL    |
| price DECIMAL(12,2)| NOT NULL    |
| capacity      INT| NOT NULL    |

```

```

| duration_minutes INT| NOT NULL      |
| description VARCHAR(1000)|      |
| active          BOOLEAN| NOT NULL, true |
| created_at      INSTANT| NOT NULL      |
| updated_at      INSTANT| NOT NULL      |
+-----+

+-----+
|          booking          |
+-----+
| id          BIGINT| PK          |
| agri_activity_id BIGINT| FK NOT NULL|
| time_slot_id  BIGINT| FK NOT NULL|
| visitor_full_name VARCHAR(150)| NN|
| visitor_email  VARCHAR(255)| NN  |
| visitor_phone  VARCHAR(30)|      |
| people_count   INT| NOT NULL   |
| payment_method VARCHAR(20)|      |
| payment_status VARCHAR(20)| UNPAID |
| status         VARCHAR(20)| PENDING |
| reminder_scheduled_at INSTANT|      |
| confirmation_sent_at INSTANT|      |
| reminder_sent_at INSTANT|      |
| created_at     INSTANT| NOT NULL  |
| updated_at     INSTANT| NOT NULL  |
+-----+

```

Table Summary

Table	Rows (design)	Purpose
`time_slot`	~500/year	Farm tour calendar slots (2/day max)
`visitor`	~2,000	Visitor database (individuals/groups)
`registration`	~4,000	Links visitor -> time slot **or** event
`briefing`	~3,000	Safety briefing per confirmed registration
`event`	~50/year	Open days, buyer visits, school/community days
`feedback`	~3,000	Satisfaction responses (tablet + survey links)
`survey_send`	~8,000	Post-visit surveys sent by email/SMS/WhatsApp
`tour_stop`	~8	Standard guided-tour stops (seeded via API)
`workshop`	~10	Tour templates / workshops (Active / Draft / Inactive)
`agri_activity`	~5	Paid agritourism offers (catalogue)
`booking`	~2,000/yr	Paid reservations linked to an activity + time slot

Enums

Enum	Values	Table Column
`TimeSlotStatus`	`AVAILABLE`, `RESERVED`, `FULL`, `CANCELLED`, `COMPLETED`	`time_slot.status`
`RegistrationStatus`	`PENDING`, `CONFIRMED`, `REJECTED`, `CHECKED_IN`, `CANCELLED`	`registration.status`
`BriefingStatus`	`PENDING`, `DONE`	`briefing.status`
`VisitorType`	`INDIVIDUAL`, `GROUP`, `SCHOOL`, `PARTNER`	`visitor.visitor_type`
`VisitPurpose`	`TOURISM`, `PURCHASE`, `PARTNERSHIP`, `INVESTMENT`, `EDUCATION`, `OTHER`	`registration.visit_purpose`
`EventType`	`OPEN_DAY`, `PARTNER_BUYER`, `SCHOOL`, `COMMUNITY`, `OTHER`	`event.event_type`
`EventStatus`	`DRAFT`, `PUBLISHED`, `CANCELLED`, `COMPLETED`	`event.status`
`FeedbackChannel`	`ON_SITE`, `EMAIL`, `SMS`, `WHATSAPP`	`feedback.origin`
`survey_send.channel`		
`SurveyStatus`	`SENT`, `RECEIVED`	`survey_send.status`
`WorkshopStatus`	`DRAFT`, `ACTIVE`, `INACTIVE`	`workshop.status`
`BookingStatus`	`PENDING`, `CONFIRMED`, `COMPLETED`, `CANCELLED`	`booking.status`
`BookingPaymentStatus`	`UNPAID`, `PAID`, `REFUNDED`	`booking.payment_status`
`BookingPaymentMethod`	`CASH`, `ORANGE_MONEY`, `MOOV_MONEY`	`booking.payment_method`

3. REST API Endpoints

Base URL: <http://localhost:8080/api/v1>

3.1 Time Slots - [/api/v1/time-slots](#)

Method	Path	Description	Status Codes
GET	/time-slots	List all slots (optional `?date=YYYY-MM-DD`)	200
GET	/time-slots/{id}	Get slot by ID	200, 404
POST	/time-slots	Create a new slot	201, 400, 409, 422
PUT	/time-slots/{id}	Update slot details	200, 404, 422
DELETE	/time-slots/{id}	Cancel a slot (soft-delete)	204, 404, 422
POST	/time-slots/{id}/assign-guide	Assign a guide (`?guideId=N`)	200, 404, 422
GET	/time-slots/availability?date=...	Availability view for a date	200

```
{
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "guideId": 5
}
```

Create/Update Request Body:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "booked": 3,
  "status": "RESERVED",
  "guideId": 5,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

TimeSlotResponse:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "remaining": 7,
  "status": "RESERVED",
  "guideId": 5
}
```

AvailabilityResponse:

3.2 Visitors - [/api/v1/visitors](#)

Method	Path	Description	Status Codes
GET	/visitors	List all visitors	200
GET	/visitors/{id}	Get visitor by ID	200, 404
POST	/visitors	Create visitor	201, 400, 409
PUT	/visitors/{id}	Update visitor	200, 404, 409

```
{
  "fullName": "Alice Dupont",
  "groupSize": 2,
  "email": "alice@example.com",
  "phone": "+241 06 12 34 56",
}
```

```

"language": "French",
"type": "INDIVIDUAL",
"specialNeeds": "Wheelchair access required"
}

```

VisitorRequest:

3.3 Registrations - /api/v1/registrations

Method	Path	Description	Status Codes
GET	/registrations	List (filter by `?timeSlotId=`, `?date=`, or `?prospect=true`)	200
GET	/registrations/{id}	Get registration by ID	200, 404
POST	/registrations	Create registration (visitor->slot, requires `visitPurpose`)	201, 400, 409, 422
PATCH	/registrations/{id}/approve	Approve -> triggers briefing creation	200, 404, 422
PATCH	/registrations/{id}/reject	Reject a pending registration	200, 404, 422
PATCH	/registrations/{id}/check-in	Check in a confirmed visitor	200, 404, 422
PATCH	/registrations/{id}/cancel	Cancel a registration	200, 404, 422

```

{
  "visitorId": 1,
  "timeSlotId": 10,
  "visitPurpose": "PURCHASE",
  "eventId": null
}

```

RegistrationRequest:

```

{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "groupSize": 2,
  "timeSlotId": 10,
  "slotDate": "2026-09-08",
  "slotStart": "09:00",
  "eventId": null,
  "visitPurpose": "PURCHASE",
  "prospect": true,
  "status": "PENDING",
  "briefingId": null,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}

```

RegistrationResponse (new fields):

3.4 Briefings - /api/v1

Method	Path	Description	Status Codes
GET	/registrations/{registrationId}/briefing	Get briefing for registration	200, 404
PATCH	/registrations/{registrationId}/briefing/deliver	Mark briefing delivered	200, 422

```

{
  "staffMember": "Guard Camille",
  "signature": "digital-signature-or-name"
}

```

BriefingDeliverRequest:

3.5 Events - /api/v1/events

Method	Path	Description	Status Codes
GET	/events	List events (`?type=`, `?date=YYYY-MM-DD`)	200
GET	/events/{id}	Get event by ID	200, 404
POST	/events	Create event (default status `DRAFT`)	201, 400, 422

```
| `PUT` | `/events/{id}` | Update event (not allowed when CANCELLED/COMPLETED) | 200, 404, 422 |
| `DELETE` | `/events/{id}` | Cancel event (soft-delete) | 204, 404, 422 |
| `POST` | `/events/{id}/publish` | Publish a DRAFT event | 200, 404, 422 |
| `GET` | `/events/{id}/registrations` | List registrations for the event | 200, 404 |
| `POST` | `/events/{id}/register` | Register a visitor for the event | 201, 400, 404, 409, 422 |
```

```
{
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall"
}
```

EventRequest:

```
{
  "id": 1,
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "booked": 3,
  "status": "PUBLISHED",
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall",
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

EventResponse:

```
{
  "visitorId": 1,
  "visitPurpose": "PURCHASE"
}
```

EventRegistrationRequest:

The registration created for an event has timeSlotId: null and carries the visitor's visitPurpose. Commercial purposes (PURCHASE, PARTNERSHIP, INVESTMENT) automatically mark the visitor as a prospect, exactly like time-slot registrations.

3.6 Feedback & Surveys - /api/v1

```
| Method | Path | Description | Status Codes |
|-----|-----|-----|-----|
| `GET` | `/feedback` | List responses (`?visitorId=`, `?from=`, `?to=`) | 200 |
| `GET` | `/feedback/summary` | Compiled stats (`?from=`, `?to=`) | 200 |
| `POST` | `/feedback` | Submit response (on-site or linked to a sent survey) | 201, 400, 404, 422 |
| `PATCH` | `/feedback/{id}/route` | Route a response to a module owner | 200, 404, 400 |
| `GET` | `/surveys` | List sent surveys (`?visitorId=`, `?status=`) | 200 |
| `POST` | `/surveys` | Send a post-visit survey (email/SMS/WhatsApp) | 201, 400, 404, 422 |
```

```
{
  "visitorId": 1,
  "surveyId": 12,
  "rating": 4,
  "briefingClear": "Yes, very clear",
  "educationalValue": "The solar tracking was interesting",
  "recommend": "Definitely",
  "comment": "Would love a German-language brochure"
}
```

FeedbackRequest: visitorId (required), surveyId (optional - links the response to a sent survey and marks it RECEIVED), rating (1-5, required), briefingClear, educationalValue, recommend, comment.

```
{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "surveyId": 12,
  "origin": "EMAIL",
  "rating": 4,
  "briefingClear": "Yes, very clear",
  "educationalValue": "The solar tracking was interesting",
  "recommend": "Definitely",
  "comment": "Would love a German-language brochure",
  "routedTo": "Sales (Mariata)",
  "submittedAt": "2026-08-26T13:00:00Z",
  "createdAt": "2026-08-26T13:00:00Z",
  "updatedAt": "2026-08-26T13:00:00Z"
}
```

FeedbackResponse:

```
{
  "averageRating": 4.2,
  "total": 18,
  "recommendPct": 92.0,
  "distribution": { "1": 0, "2": 0, "3": 3, "4": 6, "5": 9 }
}
```

FeedbackSummaryResponse (compiled weekly report):

```
{
  "visitorId": 3,
  "channel": "EMAIL",
  "messageTemplate": "Thank you for visiting Sustainable Farm! Share your feedback: [link]"
}
```

SurveyRequest:

SurveyResponse: id, visitorId, visitorName, channel, messageTemplate, sentAt, status (SENT/RECEIVED), rating (once the visitor responded), createdAt, updatedAt.

On-site tablet submissions use POST /feedback without surveyId -> origin: "ON_SITE". A sent survey becomes RECEIVED automatically when a response is linked to it.

3.7 Educational Program - /api/v1

Method	Path	Description	Status Codes
GET	/tour-stops	List guided-tour stops (ordered by position)	200
POST	/tour-stops	Create a tour stop	201, 400, 409
PUT	/tour-stops/{id}	Update a stop (position must stay unique among active stops)	200, 404, 409, 400
DELETE	/tour-stops/{id}	Deactivate a stop (soft-delete)	204, 404
GET	/workshops	List workshops/templates (?status=DRAFT ACTIVE INACTIVE)	200
POST	/workshops	Create a workshop (status DRAFT)	201, 400
PUT	/workshops/{id}	Update a workshop	200, 404, 400
POST	/workshops/{id}/publish	Publish a DRAFT workshop -> ACTIVE	200, 404, 422
POST	/workshops/{id}/deactivate	Deactivate an ACTIVE workshop -> INACTIVE	200, 404, 422

```
{
  "name": "Solar plant & tracking system",
  "position": 4,
  "description": "86.4 kwp installed, +20% yield with tracking",
  "durationMinutes": 15,
  "maxCapacity": 15,
  "location": "Solar field - observation path",
  "demo": "Live dashboard screen (Infineon sensors)",
}
```



```

"safetyNotes": "Observation path only; tracker, battery/inverter and running generator restricted"
}

```

TourStopRequest / TourStopResponse: name (required), position (required, ?1, unique among active stops), description, durationMinutes (required, ?1), maxCapacity (optional, ?1), location, demo, safetyNotes, active.

WorkshopRequest / WorkshopResponse: name (required), durationMinutes (required, ?1), targetGroup (required, e.g. All, Professional, Schools), facilitator, description, status (DRAFT by default).

Confirmed reference content (module owners: Aida, Jean-Louis, Delwende, Phares, Abdoul): standard tour stops - Welcome & safety briefing (5-10 min, Front Desk), Mango orchard (20 min, Keitt 200 trees/2 ha), Smart drip irrigation (10-15 min, max 10), Solar plant & tracking (15 min, max 15), Processing unit (20 min, max 10), Wrap-up & questions (15 min) -> max 10 pers/slot (limiting stops: irrigation + processing). Reference workshops: Standard farm tour (~100 min, All, Alix), Solar workshop (45 min, Professional, Guest Aida), Mango tasting & processing (40 min, General public, Guest Abdoul), School discovery day (120 min, Schools, P. Nikiéma).

3.8 Agritourism Booking System - /api/v1

Method	Path	Description	Status Codes
GET	/activities	List agritourism activities (alphabetical)	200
POST	/activities	Create an activity (name, price FCFA, capacity, durationMinutes)	201, 400
PUT	/activities/{id}	Update an activity	200, 404, 400
DELETE	/activities/{id}	Deactivate an activity (soft-delete)	204, 404
GET	/bookings	List bookings (?status=PENDING CONFIRMED COMPLETED CANCELLED, ?activityId=, ?date=)	200
GET	/bookings/occupancy	Occupancy % per activity	200
POST	/bookings	Create a booking (validates capacity)	201, 400, 404, 409
PUT	/bookings/{id}	Update a booking (PENDING/CONFIRMED only)	200, 404, 409, 422
POST	/bookings/{id}/pay	Record payment -> PAID	200, 404, 422
POST	/bookings/{id}/confirm	Confirm (only PENDING + PAID), sends confirmation email, schedules 24h reminder	200, 404, 422
POST	/bookings/{id}/complete	Mark a CONFIRMED booking COMPLETED	200, 404, 422
POST	/bookings/{id}/cancel	Cancel PENDING/CONFIRMED; paid -> REFUNDED; frees capacity	200, 404, 422

```

{
  "name": "Solar workshop",
  "price": 5000,
  "capacity": 15,
  "durationMinutes": 45,
  "description": "Guided visit of the 86.4 kwp solar plant with tracking"
}

```

AgriActivityRequest / AgriActivityResponse: name (150), price (BigDecimal ? 0, FCFA), capacity (? 1), durationMinutes (? 1), description (1000), active.

```

{
  "activityId": 1,
  "timeSlotId": 2,
  "visitorFullName": "Lucas Weber",
  "visitorEmail": "lucas@example.com",
  "visitorPhone": "+22612345678",
  "peopleCount": 5,
  "paymentMethod": "ORANGE_MONEY"
}

```

BookingRequest / BookingResponse: activityId, timeSlotId, visitorFullName, visitorEmail (valid email), visitorPhone, peopleCount (? 1), paymentMethod (CASH, ORANGE_MONEY, MOOV_MONEY). Response adds reference (BK-00001 format), activityName, activityPrice, slot date/times, totalAmount (price × people), paymentStatus, status, reminderScheduledAt.

Reference data from the mockup: Standard tour 5,000 FCFA/person (family rate 3,500 FCFA); payment accepted in cash, Orange Money, Moov Money.

Emails (confirmation + rappel 24 h). À la confirmation, un email de confirmation est envoyé au visiteur (référence BK-xxxxx, activité, date/créneau, personnes, total) et confirmationSentAt est posé. Un job planifié (BookingReminderJob, toutes les 60 s par défaut) envoie le rappel « moins de 24 h » aux réservations confirmées dont l'échéance est atteinte (reminderScheduledAt ? now) et passe reminderSentAt. Un échec d'envoi ne bloque jamais la réservation : l'erreur est loggée et la confirmation est maintenue. Deux implémentations de NotificationService :

Mode	Condition	Comportement
ConsoleNotificationServiceImpl	app.mail.enabled=false (défaut)	Email loggé en console (dev)
SmtplibNotificationServiceImpl	app.mail.enabled=true	Envoi réel via JavaMailSender (SMTP)

4. Business Rules

Scheduling Rules (SchedulingRules.java)

Rule	Value	Source
Max visitors per slot	**10**	Confirmed with module owners (Abdoul)
Max slots per day	**2**	Mockup: morning + afternoon
Closed day	**Sunday**	All module owners
Morning slot	09:00-11:00	Default
Afternoon slot	14:00-16:00	Default

Registration Rules

Rule	Enforcement
Cannot register on a **CANCELLED** slot	BusinessRuleException
No duplicate registration (visitor + slot)	ConflictException (409)
Group size must fit remaining capacity	BusinessRuleException
Only **PENDING** -> approve/reject	BusinessRuleException
Only **CONFIRMED** -> check-in	BusinessRuleException
Cannot cancel after **CHECKED_IN**	BusinessRuleException
CONFIRMED -> auto-creates Briefing (PENDING)	Service layer
Briefing delivery requires PENDING status	BusinessRuleException

Event Rules

Rule	Enforcement
endDateTime must be after startDateTime	BusinessRuleException
Only **DRAFT** events can be published	BusinessRuleException
Cannot publish an event that has already ended	BusinessRuleException
Cannot update **CANCELLED** or **COMPLETED** events	BusinessRuleException
Cannot cancel a **COMPLETED** event	BusinessRuleException
Registration only on **PUBLISHED** events	BusinessRuleException
No duplicate registration (visitor + event)	ConflictException (409)
booked + groupSize > maxCapacity -> reject	BusinessRuleException (422)
Booked count excludes REJECTED / CANCELLED registrations	Repository query

Feedback & Survey Rules

Rule	Enforcement
Rating must be between 1 and 5	Bean validation (400)
Response linked to a survey must match the survey's visitor	BusinessRuleException (422)
A survey can only be answered once (RECEIVED is final)	BusinessRuleException (422)
Survey channel cannot be ON_SITE (tablet responses are direct)	BusinessRuleException (422)
Response without a survey -> recorded as ON_SITE	Service layer
Linking a response marks the survey RECEIVED	Service layer
Routing targets are free-text tags (no coupling to other modules)	Data model

Educational Program Rules

Rule	Enforcement
-----	-----
Position must be ? 1	Bean validation (400)
Position must be unique among active stops	`ConflictException` (409)
Deactivating a stop frees its position (soft-delete)	`DELETE /tour-stops/{id}`
Create -> status `DRAFT`	Service layer
Only DRAFT workshops can be published	`BusinessRuleException` (422)
Only ACTIVE workshops can be deactivated	`BusinessRuleException` (422)

Agritourism Booking Rules

Rule	Enforcement
-----	-----
Capacity counts non-cancelled bookings only	Repository aggregate query
A booking can never exceed the slot capacity (10 pers)	`ConflictException` (409)
A booking can never exceed the activity capacity	`ConflictException` (409)
Bookings only on active activities and non-cancelled slots	`BusinessRuleException` (422)
Cannot update COMPLETED / CANCELLED bookings	`BusinessRuleException` (422)
Can only confirm a PENDING booking that is PAID	`BusinessRuleException` (422)
Confirm schedules the 24 h reminder (slot start ? 24h)	Service layer
Completing requires CONFIRMED status	`BusinessRuleException` (422)
Cancelling a PAID booking marks it REFUNDED	Service layer
Confirm sends the booking confirmation email	`NotificationService` (never blocks)
24 h reminder sent for due, unsent reminders	`BookingReminderJob` (fixed delay, 60 s)
Reminder / confirmation deliveries are idempotent	(`confirmationSentAt`, `reminderSentAt`)
Repository query + service	

Visitor Rules

Rule	Enforcement
-----	-----
Duplicate email ->	`ConflictException` (409) On create and update
`@NotBlank` on fullName	Bean validation (400)
`@Email` on email	Bean validation (400)

Lead Generation (Prospect Detection)

The visitor management app doubles as a marketing tool for the Sales & Marketing team. When a visitor registers with a commercial purpose, the system automatically flags them as a prospect.

How it works:

1. During registration, the visitor must select a visitPurpose (mandatory field)
2. If the purpose is PURCHASE, PARTNERSHIP, or INVESTMENT, the system sets isProspect = true automatically
3. Sales & Marketing can query all prospects via GET /api/v1/registrations?prospect=true

VisitPurpose values:

Purpose	Prospect?	Description
-----	-----	-----
`TOURISM`	No	Leisure visit, tourism
`PURCHASE`	Yes	Interested in buying products
`PARTNERSHIP`	Yes	Business partnership opportunity
`INVESTMENT`	Yes	Investment opportunity
`EDUCATION`	No	Training, educational visit
`OTHER`	No	Other reasons

GET /api/v1/registrations?prospect=true

Prospect query example: Returns all registrations where isProspect = true, including visitor name, email, phone, and visit purpose.

5. Status Flow Diagrams

```

+-----+
create  | AVAILABLE|
-----> +-----+
      |
      +-----?-----+
      |         |         |
register assign  cancel
      |   guide   |         |
      v           |         v
+-----+ | +-----+
| RESERVED | | | CANCELLED|
+-----+ | +-----+
      |   |         |
full |   |         |
      v   |         |
+-----+ |         |
| FULL |<-----+ (booked >= capacity)
+-----+
      |
end_time passes
      v
+-----+
| COMPLETED |
+-----+

```

```

+-----+
register | PENDING |
-----> +-----+
        |
        +-----?-----+
        |       |       |
    approve reject  cancel
        |       |       |
        v       v       v
+-----+ +-----+ +-----+
|CONFIRMED| |REJECTED| |CANCELLED|
+-----+ +-----+ +-----+
        |
    check-in
        |
        v
+-----+
|CHECKED_IN|
+-----+
        |
    cancel blocked
        |
    (final state)

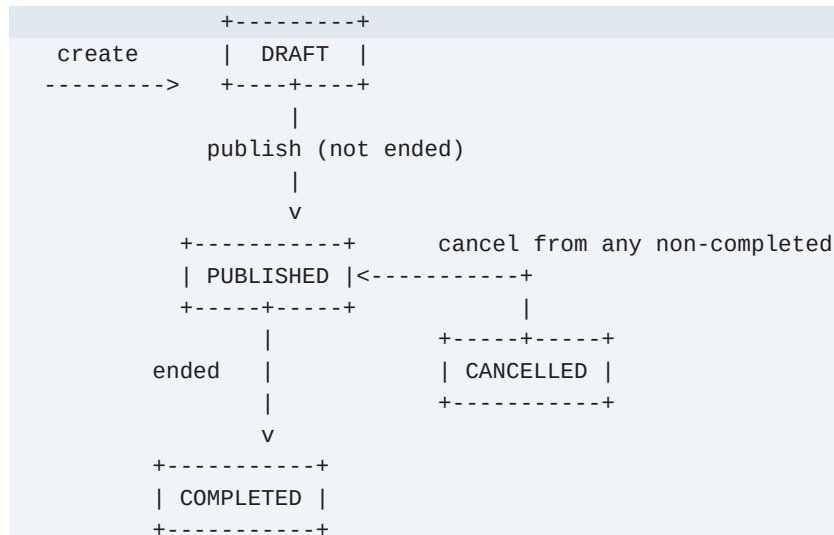
```

```

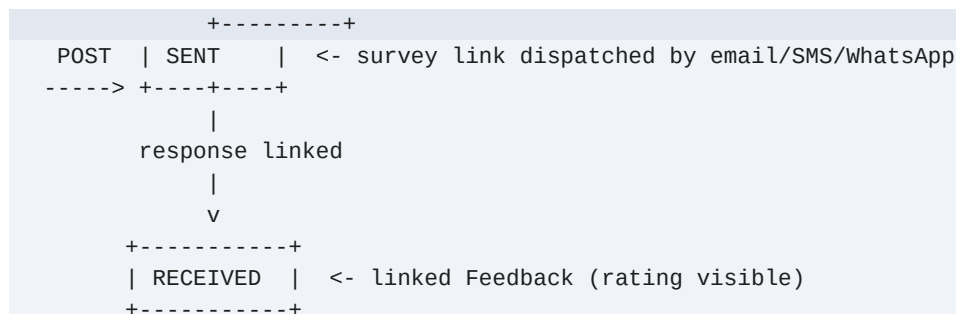
+-----+
auto | PENDING | <- created when registration CONFIRMED
-----> +-----+
      |
      deliver briefing
      |
      v
+-----+
|  DONE  | <- staffMember + signature + deliveredAt
+-----+

```

Event Status Flow



Survey Status Flow



6. Error Handling

All errors return a consistent JSON envelope via `GlobalExceptionHandler`:

```
{
  "status": 404,
  "error": "Not Found",
  "message": "Time slot 999 not found",
  "path": "/api/v1/time-slots/999"
}
```

Exception	HTTP Status	When
<code>ResourceNotFoundException</code>	404	Entity not found
<code>ConflictException</code>	409	Duplicate email, duplicate registration, overlapping slot
<code>BusinessRuleException</code>	422	Invalid state transition, capacity exceeded, Sunday creation
<code>MethodArgumentNotValidException</code>	400	Bean validation failures (<code>@NotBlank</code> , <code>@Email</code> , etc.)

7. Test Coverage Summary

Layer	Annotation	Tests	Framework
Repository	<code>@DataJpaTest</code>	+ H2 39	Spring Data + AssertJ
Service	<code>@ExtendWith(MockitoExtension.class)</code>	94	Mockito + AssertJ
Controller	<code>@WebMvcTest</code>	+ MockMvc 82	MockMvc + Mockito
Context	<code>@SpringBootTest</code>	4	Spring Boot (+MockMvc CORS preflight)
Core (notification)		plain Mockito 3	Mockito + AssertJ
Total		**222**	

Test Files

```

src/test/java/com/infineonbit/sustainablefarm/
+-- SustainableFarmApplicationTests.java
+-- modules/visitormanagement/
    +-- repository/
        | +-- TimeSlotRepositoryTest.java      (5 tests)
        | +-- VisitorRepositoryTest.java       (4 tests)
        | +-- RegistrationRepositoryTest.java  (5 tests)
        | +-- BriefingRepositoryTest.java      (2 tests)
        | +-- EventRepositoryTest.java         (4 tests)
        | +-- FeedbackRepositoryTest.java      (7 tests)
        | +-- EducationalProgramRepositoryTest.java (5 tests)
        | +-- AgriTourismRepositoryTest.java (7 tests)
    +-- service/
        | +-- SchedulingServiceImplTest.java   (13 tests)
        | +-- RegistrationServiceImplTest.java (21 tests)
        | +-- EventServiceImplTest.java        (16 tests)
        | +-- FeedbackServiceImplTest.java     (12 tests)
        | +-- EducationalProgramServiceImplTest.java (12 tests)
        | +-- BookingServiceImplTest.java      (18 tests)
        | +-- BookingReminderJobTest.java      (2 tests)
    +-- controller/
        +-- TimeSlotControllerTest.java       (9 tests)
        +-- VisitorControllerTest.java        (7 tests)
        +-- RegistrationControllerTest.java    (12 tests)
        +-- BriefingControllerTest.java       (5 tests)
        +-- EventControllerTest.java          (13 tests)
        +-- FeedbackControllerTest.java       (10 tests)
        +-- EducationalProgramControllerTest.java (12 tests)
        +-- BookingControllerTest.java        (13 tests)
+-- core/notification/
    | +-- NotificationServiceTest.java        (3 tests)
+-- CorsConfigurationTest.java               (3 tests, @SpringBootTest + MockMvc)

```

8. Running the Backend

```

# Start PostgreSQL + Backend
docker compose up -d backend

# Backend available at
http://localhost:8080

# Swagger UI
http://localhost:8080/swagger-ui/index.html

# Actuator health
http://localhost:8080/actuator/health

# Run tests (Docker, no local JDK required)
docker run --rm -v "$PWD":/src -w /src \
    -v "$HOME/.m2":/root/.m2 \
    maven:3.9-eclipse-temurin-21 ./mvnw test

# Compile only
docker run --rm -v "$PWD":/src -w /src \
    -v "$HOME/.m2":/root/.m2 \
    maven:3.9-eclipse-temurin-21 ./mvnw compile

```

Email notifications (env vars)

Real SMTP delivery requires MAIL_ENABLED=true plus SMTP credentials. Without it, emails are only logged to the console (dev mode).

Variable	Default	Description
-----	-----	-----
`MAIL_ENABLED`	`false`	`true` -> real SMTP, `false` -> console log
`MAIL_HOST`	*(empty)*	SMTP host (e.g. `smtp.orange.bf`)
`MAIL_PORT`	`587`	SMTP port
`MAIL_USERNAME`	*(empty)*	SMTP username
`MAIL_PASSWORD`	*(empty)*	SMTP password
`MAIL_FROM`	`visits@sustainable-farm.local`	Sender address
`MAIL_REMINDER_INTERVAL_MS`	`60000`	Reminder poll interval (fixed delay)

CORS (env vars)

The backend allows cross-origin calls on /api/ for the configured origins. Adjust APP_CORS_ORIGINS to your frontend addresses (Vite dev server / container).

Variable	Default	Description
-----	-----	-----
`APP_CORS_ORIGINS`	`http://localhost:3000,http://localhost:5173`	`Comma-separated allowed origins`

Dev reference data (env vars)

Enable with SPRING_PROFILES_ACTIVE=dev. DevDataSeeder (idempotent: seeds each table only when empty) loads the confirmed mockup data - never active in production:

- Tour stops: 6 stops (Accueil -> Mango orchard -> Irrigation -> Solar -> Processing -> Wrap-up)
- Workshops: 4 (Standard farm tour, Solar workshop, Mango tasting [DRAFT], School day)
- Agri activities: 3 (Standard tour 5 000 FCFA, Mango tasting 3 000 FCFA, Solar workshop 5 000 FCFA)
- Time slots: next 7 days (2/day 09-11 & 14-16, Sundays skipped, capacity 10)
- Event: "Journée portes ouvertes" (OPEN_DAY, PUBLISHED, +2 weeks)
- Sample visitor: "Awa Ouédraogo" (group of 8) with a pending PURCHASE registration (prospect)

Variable	Default	Description
-----	-----	-----
`SPRING_PROFILES_ACTIVE`	*(empty)*	`dev` -> seed reference data

9. Technology Stack

Component	Version	Notes
-----	-----	-----
Java	21	LTS
Spring Boot	4.1.0	Modular monolith
Spring Data JPA	4.1.0	Derived queries + `@Query`
Hibernate ORM	7.4.1	`ddl-auto=update` (no Flyway)
PostgreSQL	17-alpine	Production database
H2	2.x	Test-only in-memory DB
SpringDoc OpenAPI	2.8.5	Swagger UI auto-generated
Spring Mail	4.1.0	`JavaMailSender` + SMTP (emails booking)
Lombok	1.18.46	Boilerplate reduction
Jackson	3.1.4	JSON serialization
Mockito	5.23.0	Service + controller tests
AssertJ	3.27.7	Fluent assertions

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